higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 A fillet joint is triangular in shape✓ and it is used to weld materials at right angles to one another where two flat metal plates are placed in the shape of a T-bar.✓ The joint does not leave a gap between the two plates.✓ (3)

1.2 The members to be welded together form a V-shape.✓ The weld material must fill the V-cavity.✓ (2)

1.3 Penetration is the depth of the weld through the joint.✓ A weld can penetrate all the way through the joint and it is called a full penetration.✓ Joint penetration is the distance from the middle of the weld face to the lowest point of the weld material.✓ Root penetration is the distance from the root of the weld face to the lowest point in the joint, where there is weld material.✓
(Any 3 × 1) (3)

1.4 The weld face is the surface of the weld that is not in direct contact with the base metals. (1)

1.5 Weld length refers to the length of the line made by the weld. (1)
[10]

QUESTION 2

2.1
A Included angle
B Pitch centres of weld
C Weld length
D Weld symbol (fillet weld)
E Weld size
F Reference line
G Arrow pointing at weld
H Weld all around
I Site weld
J Root penetration
(10 × 1) (10)

2.2
- Incorrect welding technique
- Incorrect setting of the welding machine
- Arc length too long
- Arc force setting too high (4)

2.3
- Check if complete penetration has been obtained.
- Check that the texture of the metal is consistent and smooth.
- Check for splatter on the work piece.
- The dimensions of the weld must be the same as those on the drawing.
- Check if complete fusion has been obtained between the base metal and the filler metal.
- Check for any visual defects. (6)

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2.4
- Ensure the welding equipment is clean and defect free.
- Ensure the welding equipment is neatly stored. (2)

2.5
- The supervisor must know about faulty tools.
- Using defective tools places you in harm's way.
- Faulty tools must be repaired or replaced. (3)

[25]

QUESTION 3

3.1 3.1.1 When cylinders are not in use, remove its regulator and place a valve cap on it.✓ All cylinders should be secured tightly in an upright position to prevent them from falling over.✓ (2)

3.1.2 Ensure that flashback arrestors are installed on the regulator outlets.✓ Ensure that the check valves are installed on the torch inlets.✓ Follow the correct start-up procedures.✓ Keep acetylene pressure below 1 bar.✓ (4)

3.1.3 Remove all sources of oil and grease from the weld area.
Ensure that there is no oil or grease on your clothes and PPE. (2)

3.1.4 Remove all flammable material from the welding area.✓ Always keep a fire extinguisher near the welding area.✓ (2)

3.2 A – Oxidising flame
B – Neutral flame
C – Carburising flame (3)

3.3
- Close the acetylene valve on the torch completely. It is very important to close the acetylene first and then the oxygen.
- Close the oxygen valve on the torch completely.
- Close the valve on the acetylene cylinder completely.
- Open the acetylene valve on the torch again to bleed the acetylene from the hose.
- When no more gas blows from the torch, close the acetylene valve on the torch.
- Open the acetylene regulator valve.
- Repeat the process for the oxygen cylinder. (7)

[20]

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QUESTION 4

4.1
4.1.1 False
4.1.2 True
4.1.3 False
4.1.4 True
4.1.5 False
4.1.6 True
4.1.7 False
4.1.8 True
4.1.9 False
4.1.10 True

(10 × 1) (10)

4.2
- Aluminium dust can catch fire easily, especially when mixed with steel dust.
- The vapour produced while welding can be hazardous if inhaled, as it can cause memory loss.
- The molten aluminium, if mixed with water, chemical substances or contaminants, can cause an explosion.
- If the molten aluminium comes into contact with any part of your body, it can cause serious burns.
- When molten aluminium splashes, precautions must be taken by the welder to wear eye and face protection as well as protective clothing. (5)

4.3
4.3.1
- Must not be stored in direct sunlight as the cylinders are under pressure
- Must be stored in an upright position and fastened to the wall

4.3.2
- Store in dry conditions.
- Avoid contact with water.

4.3.3
- Plastic bags can be used for storage to prevent surface contamination.
- Store in places that are free of dust, scale, oil, grease and moisture.

4.3.4
- Very brittle due to heating and cooling during welding
- The torch and nozzle must not be dropped or knocked against other objects.

4.3.5
- Should be loosely coiled after use
- Should be free from dust, oil, grease and scale

(2 × 5) (10)
[25]

QUESTION 5

5.1
A Shielded gas cylinder
B Welding torch
C Wire feeder
D Work piece lead (earth)
E Welding wire
F Grounding
G Power line
H Welding power source
I Gas hose
J Gas flow adjuster

(10 × 1) (10)

5.2 Amperage = $\frac{\text{Material thickness}}{0,0254}$ ✓✓

$= \frac{12}{0.0254}$ ✓✓

= 472,44 A✓ (5)

5.3
- Protractor
- Set square
- Spirit level (3)

5.4
- Push technique
- Pull technique (2)

[20]

TOTAL: 100

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